MIL-H-10330A
21 SEPTEMBER 1950
SUPERSEDING
MIL-H-10330(CmlC)
31 May 1950

MILITARY SPECIFICATION

HANSA YELLOW G

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one grade of hansa yellow G for use as a component of M5 liquid vesicant detector paint.

2. APPLICABLE SPECIFICATIONS, AND OTHER PUBLICATIONS

2.1 Specifications.—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATIONS

RR-S-366—Sieves; Standard, Testing.

TT-P-141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

U. S. ARMY SPECIFICATION

100-2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

(*Army.*—Copies of specifications may be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal, Military, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Federal, Military, and U. S. Army specifications may be obtained upon application to the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. — The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Center, Norfolk 11, Va.)

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

[1] Applicable only to Navy purchases.

905079—50—(11-198)

FSC 6665

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:56Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Mass color.—The mass color of the dye shall be not less than that of the standard sample which will be furnished by the interested service (see 6.2).

3.2 Tinting strength.—The tinting strength shall be not less than that of the standard sample.

3.3 Particle size.—The material shall be in such a state of subdivision that not less than 98 percent of the material shall pass through a No. 200 U.S. Standard sieve conforming to Specification RR–S–366.

3.4 Volatile matter.—The volatile matter shall be not greater than 5.0 percent.

3.5 Ash.—The material shall yield not more than 3.0 percent of ash upon ignition.

3.6 Total chlorine.—The total chlorine content shall be not more than 0.2 percent.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—A lot shall consist of the hansa yellow G produced by one manufacturer with no change in process or materials in no more than 24 consecutive hours, providing the operation is continuous. In the event the process is a batch operation, each batch shall consitute a lot. (See 6.3.)

4.2 Sampling.—Ten containers shall be selected, at random, from each lot for test purposes. A representative specimen shall be taken from each container selected. The ten specimens shall be divided into two groups of five specimens each and a composite sample prepared from each group. If there are fewer than ten containers in the lot, specimens shall be taken from each container and composite samples formed in the same manner. Place each composite specimen in a clean, dry container, seal, and label in a manner that will identify the specimen with the containers and lot represented.

4.3 Inspection of packing and marking.—The inspector shall ascertain that the packing and marking of the material conform to section 5 of this specification.

4.4 Tests.

4.4.1 *Reagents.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, and corrections shall be applied when significant.

4.4.2 *Physical.*—The tests for the physical requirements shall be conducted in accordance with Specification TT–P–141 as shown in table I.

TABLE I.—*Physical tests*

Requirement	Method
Mass color	421.1
Tinting strength	422.1[a]
Particle size	410.1[b]
Volatile matter	407.1[c]
Ash	526.1

[a] Follow procedure specified for C.P. toluidine toner.
[b] Use a No. 200 U. S. Standard sieve.
[c] Heat 2 hours.

4.4.3 *Total chlorine.*—Total chlorine shall be determined as follows: Weigh to the nearest mg. approximately 0.7 gm. of the sample and transfer to a Parr sulfur bomb cup fitted with a lead gasket. Add approximately 0.5 gm. of sodium hydroxide pellets and 10 gm. of sodium peroxide-sucrose mixture (12 gm. of 40 mesh sucrose to 250 gm. of sodium peroxide). Assemble and tighten the bomb and place in a holder within a shield. (A satisfactory shield for use with the Parr-bomb ignition may be made from ⅛-inch sheet iron, approximately 15 inches high by 10 inches wide by 10 inches deep, with one 10- by 15-inch side open. The shield should have an interior shelf, located approximately 10 inches above the bottom, provided with a

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:56Z
Check the source to verify that this is the current version before use.

hexagonal opening for holding the bomb during ignition in an approximately central position with reference to the floor of the shield.) Ignite the contents of the bomb by directing the flame of an oxy-gas burner against the bottom of the cup. After the initial ignition, continue the heating until the lower half of the cup attains an even, cherry-red color. Remove the bomb from the shield, allow to stand for approximately five minutes, and cool in water. Remove the outer fastenings, without disturbing the cap, and wash the exterior of the cup and cap with water, discarding the washings. Remove the cap, place the cup on its side in an 800-mil. beaker, wash the material adhering to the underside of the cap into the beaker with 1:20 nitric acid, cover with a watch glass, and add water until the cup is approximately three quarters covered. When the material in the cup is apparently dissolved, add approximately 2 gm. of sodium peroxide, stir, boil until the peroxides are decomposed and allow to cool. Wash the cup in 1:20 nitric acid, add 1:1 nitric acid in small portions, allowing sufficient time between additions of the acid for the completion of the reaction, in order to avoid a large excess of acid. When the solution is clear, allow to cool, and determine the total chlorine by the Volhard method. (The iron in solution acts as indicator.)

$$\text{Percent Cl} = \frac{3.55\ (VN - V_1 N_1)}{W}$$

Where

V = ml. $AgNO_3$
N = normality of $AgNO_3$
V_1 = ml. KCNS
N_1 = normality of KCNS
W = gm. of sample.

4.4.4 *Rejection and resubmission.* — If either composite specimen fails to conform to this specification, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made of each container in the lot and removing the nonconforming drums at no expense to the Government. The contractor may then resubmit the remaining portion of the lot for acceptance testing.

5. PREPARATION FOR DELIVERY

5.1 Packing.—The material shall be packed either in removable head steel drums conforming to Specification 37D of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., or in fiber drums conforming to Specification 21A, of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc. for an authorized 200 lb. net weight.

5.2 Marking.—In addition to any special marking required by the contract or order shipping containers for the Army shall be marked in accordance with Specification 100-2; for the Navy, in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the title, and date of this specification.

6.2 The first satisfactory shipment of the dye from a manufacturer will be considered a permanent standard for that manufacturer.

6.3 Batch. — A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Inspection.—Inspection will not necessarily include all drawing and specification requirements; however, the omission of inspection does not constitute a waiver of that requirement.

6.5 Sampling and testing.—If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government inspector, at his discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right

Source: https://assist.dla.mil -- Downloaded: 202640-17T13:56Z
Check the source to verify that this is the current version before use.

to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Chemical Corps

Other interest:
Navy—OSh.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:56Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
EDGEWOOD ARSENAL
ATTN: SMUEA-QA-WS-S
EDGEWOOD ARSENAL, MARYLAND
21010

FOLD

PLATE NO. 19419 (BACK)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:56Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).*

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:56Z
Check the source to verify that this is the current version before use.